

CREDIT CARD FRAUD DETECTION

A PROJECT REPORT

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Under the supervision of

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Submitted to

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DECLARATION

We hereby declare that the work presented in this report entitled “**Credit Card Fraud Detection System**” was carried out by us. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

We have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments, results, that are not my original contribution. We have used quotation marks to identify verbatim sentences and given credit to the original authors/sources.

We affirm that no portion of my work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, we shall be fully responsible and answerable.

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CERTIFICATE

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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ABSTRACT

In the digital economy, the detection and prevention of credit card fraud have become crucial challenges due to the increasing frequency and complexity of online transactions. This project presents a hybrid approach to credit card fraud detection by employing both supervised and unsupervised machine learning techniques—namely, Logistic Regression and Isolation Forest. The system is developed using a real-world dataset of European credit card transactions, characterized by severe class imbalance and anonymized features.

In the evolving landscape of the digital economy, **credit card fraud** remains one of the most pressing challenges for financial institutions and e-commerce platforms. With the exponential growth in online transactions, fraudsters have adopted increasingly sophisticated techniques, making fraud detection both more complex and more critical. This project addresses the issue by proposing a **hybrid machine learning-based system** that leverages the strengths of both **supervised** and **unsupervised learning** paradigms to enhance the accuracy and adaptability of fraud detection mechanisms.

Logistic Regression is utilized for binary classification after balancing the dataset through undersampling, enabling effective identification of fraudulent transactions based on historical patterns. In parallel, the Isolation Forest algorithm is applied to detect anomalies within the entire dataset without requiring class labels, making it suitable for discovering novel fraud behaviors. The system is evaluated through accuracy metrics, confusion matrices, and visualizations, demonstrating its ability to distinguish between normal and suspicious activities.

The results from this hybrid approach indicate that combining both supervised and unsupervised techniques offers a **robust, flexible, and scalable solution** to the problem of credit card fraud detection. While Logistic Regression excels in identifying known fraud behaviors, the Isolation Forest algorithm complements it by detecting emerging or uncommon fraudulent patterns without relying on labeled data.

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INTRODUCTION

1.1. Overview

Credit card fraud detection is a fundamental pillar of modern financial cybersecurity, acting as a frontline defence mechanism to protect both consumers and institutions from significant monetary loss and reputational damage. As digital transformation reshapes the global economy, the widespread adoption of online payment systems, mobile banking applications, contactless payments, and e-commerce platforms has led to a dramatic surge in credit card usage. This exponential growth in digital transactions, while beneficial for convenience and scalability, has simultaneously opened new avenues for malicious actors to exploit vulnerabilities within financial ecosystems.

Credit card fraud involves the unauthorized use of payment card data to conduct financial transactions, either through stolen physical cards or compromised digital information. Fraud can manifest in various forms, including card-not-present (CNP) fraud, which occurs in online environments without physical cards; lost or stolen card fraud, where unauthorized users gain access to a card; counterfeit fraud involving cloned cards; account takeover fraud, which exploits compromised credentials; and application fraud, where attackers create fake identities to open fraudulent accounts. These evolving attack vectors make it increasingly difficult for conventional fraud prevention systems to keep pace.

Traditional fraud detection systems, largely based on rule-based engines and manually curated thresholds, follow a deterministic approach. They rely on predefined parameters such as geographic anomalies, large transaction amounts, unusual merchant types, or excessive transaction frequency to flag potential fraud. While useful for capturing well-known patterns, these systems are inherently rigid, often unable to adapt to new fraud tactics and behaviours. This rigidity leads to high false-positive rates—legitimate transactions being mistakenly identified as fraudulent—resulting in declined transactions, customer dissatisfaction, and loss of business opportunities. Moreover, such systems are reactive rather than proactive, detecting fraud only after it has occurred or during post-transaction analysis.

To address these limitations, this project proposes a **machine learning-based Credit Card Fraud Detection system** that leverages both supervised and unsupervised learning techniques. By shifting from a rules-based to a data-driven approach, machine learning models can generalize from historical data,

identify non-obvious correlations, and dynamically evolve with new fraud trends. This enables proactive fraud prevention and improved decision-making with minimal human intervention.

The project utilizes a real-world dataset comprising over 284,000 anonymized credit card transactions, with only 492 instances of fraud—highlighting a highly imbalanced class distribution typical of real-world fraud scenarios. This imbalance presents a significant challenge, which is addressed using strategic sampling and data balancing techniques such as under sampling or Synthetic Minority Over-sampling Technique (SMOTE). These approaches ensure that the models do not become biased towards the majority class (legitimate transactions) and maintain sensitivity to minority class patterns (fraud).

Two machine learning techniques form the core of this system:

- **Logistic Regression:** A probabilistic, interpretable supervised learning model used for binary classification. It learns decision boundaries based on historical labelled data, calculating the likelihood that a transaction is fraudulent based on its features. Its simplicity, efficiency, and ability to provide probabilities make it ideal for high-speed fraud prediction in real-time systems.
- **Isolation Forest:** An unsupervised anomaly detection algorithm that isolates rare and suspicious data points by randomly selecting features and split values. Unlike supervised models, it does not require labelled training data, making it particularly suitable for identifying novel fraud attempts that deviate from normal patterns. It's efficient on large datasets and effective for highlighting outliers.

The intelligent system built here processes a wide array of transaction features—including time of transaction, amount, and anonymized behavioural attributes (V1-V28)—to detect fraud patterns. The data pipeline involves exploratory data analysis (EDA), feature scaling, model training, performance evaluation, and visualization. Confusion matrices, classification reports, and scatter plots provide insight into model performance, helping to optimize thresholds and understand fraud distribution visually.

Key outcomes of this machine learning-driven system include:

- **Improved Fraud Detection Accuracy:** By uncovering subtle patterns in the data, the models enhance the precision and recall of fraud classification.
- **Reduction in False Positives:** Through more nuanced decision boundaries and anomaly scoring, legitimate transactions are less likely to be wrongly flagged.

- **Dynamic Adaptability:** The models can be retrained on recent data, allowing them to evolve in response to new attack vectors without manual updates.
- **Real-Time Capability:** Efficient algorithms and optimized inference pipelines enable rapid decision-making, essential for time-sensitive fraud prevention.

Additionally, the system's modular design allows it to be deployed as part of a real-time transaction monitoring service using tools like Flask or Streamlit. It can be integrated with financial APIs, connected to customer relationship management (CRM) systems, or extended with deep learning models such as LSTM (for sequential analysis) or autoencoders (for complex anomaly detection).

Ultimately, the goal of this project is not only to detect fraud but also to build an adaptable framework that contributes to **digital trust and financial resilience**. As attackers become more sophisticated, so too must the defences. Machine learning offers a sustainable, intelligent alternative to traditional methods, capable of learning from data, adapting to emerging threats, and scaling with system growth.

In conclusion, this project exemplifies how the intersection of artificial intelligence and finance can yield practical, high-impact solutions. It demonstrates the viability of machine learning models like Logistic Regression and Isolation Forest in mitigating fraud risks, protecting consumer interests, and preserving institutional integrity. As cybercrime continues to evolve, the adoption of such intelligent systems will be vital in shaping a secure, transparent, and resilient digital economy.

1.2. Project goal

The primary objective of this project is to develop a robust and efficient system for the detection of **fraudulent credit card transactions**. In today's digitally driven financial environment, fraud detection plays a critical role in protecting consumers from unauthorized charges, safeguarding personal and financial data, and ensuring the integrity of financial institutions. Accurately identifying fraudulent activity is essential, as failure to do so can lead to severe financial losses, erosion of customer trust, and potential regulatory consequences. This project is focused on leveraging the power of **machine learning (ML) algorithms** to automatically distinguish between legitimate and suspicious transactions within a credit card dataset.

To achieve this, the project employs multiple machine learning approaches—including both **supervised** and **unsupervised techniques**—that are capable of learning from historical transaction data to predict whether a given transaction is fraudulent. Among the techniques used are **Logistic Regression**, which is known for its simplicity and effectiveness in binary classification tasks, and **Isolation Forest**, a widely used anomaly detection method that excels in identifying rare and abnormal patterns without the need for labelled data. These models are trained, validated, and tested on a real-world dataset to evaluate their ability to generalize and correctly classify unseen transactions.

A significant aspect of this project involves the **comparative analysis of model performance**. The results generated by each machine learning method are evaluated using a range of performance metrics such as **accuracy**, **precision**, **recall**, **F1-score**, and **confusion matrices**. Additionally, graphical representations—such as ROC curves, bar charts, and scatter plots—are included to visually convey the model outcomes and assist in identifying the most effective technique. This comparative evaluation helps in understanding the trade-offs between different models, such as interpretability versus predictive power or precision versus recall, which is especially important in the context of fraud detection where minimizing false negatives is crucial.

Beyond model implementation, the project also involves a **review of existing literature and methodologies** to understand the various techniques that have been previously applied to fraud detection problems. This includes a study of traditional rule-based systems, statistical methods, and recent advances in artificial intelligence and machine learning. By exploring past research, the project builds on established knowledge and identifies gaps or challenges that can be addressed through novel model design or data preprocessing strategies.

In summary, this project aims not only to detect fraudulent credit card transactions using advanced machine learning techniques but also to provide a holistic understanding of the field by analysing, comparing, and visualizing the outcomes of multiple models. The end goal is to determine the most accurate and efficient approach for real-world application, ensuring both consumer protection and financial security in increasingly digital financial ecosystems.

LITERATURE REVIEW

Over the past several years, credit card fraud detection has emerged as a critical area of focus across multiple domains, including machine learning, cybersecurity, data science, and financial technology. This increasing attention stems from the exponential rise in digital payment adoption and the parallel escalation in fraudulent activities that exploit system vulnerabilities. As a result, researchers have dedicated significant efforts to developing advanced analytical models that can accurately and efficiently detect fraud while minimizing disruption to legitimate transactions. These efforts have led to the evolution of various detection methodologies, broadly categorized into **supervised learning**, **unsupervised learning**, and **hybrid systems** that integrate both paradigms.

Supervised learning techniques have been widely adopted due to the availability of labelled transaction data. In one widely referenced study using the **European credit card fraud dataset**—which contains anonymized transaction features and highly imbalanced labels—multiple algorithms were benchmarked for their predictive accuracy. The models tested included **Logistic Regression (LR)**, **Decision Trees (DT)**, and **Random Forest (RF)**. The results demonstrated that Random Forest achieved the highest classification accuracy at **95.5%**, followed closely by Decision Trees at **94.3%**, and Logistic Regression at **90%**. The superior performance of RF was attributed to its ensemble nature, which combines the outputs of multiple decision trees to enhance robustness and reduce overfitting. Moreover, RF is inherently capable of handling **non-linear interactions and high-dimensional feature spaces**, making it well-suited for complex fraud detection tasks.

Beyond traditional classifiers, further research has emphasized the role of **k-Nearest Neighbors (KNN)** and **outlier detection techniques**. These methods are particularly effective when labelled data is scarce or incomplete—a common challenge in fraud detection where not all fraudulent behaviours are captured. Studies referenced in [2] and [3] showed that KNN could identify anomalies by comparing new transactions to historical patterns within a defined local neighborhood. This proximity-based approach allows for the detection of subtle deviations that may not be apparent to global models. However, KNN's limitations in scalability—especially in real-time applications involving large-scale datasets—restrict its utility in production environments unless optimized through techniques like approximate nearest neighbors or dimensionality reduction.

In **paper [4]**, a broader experimental comparison of classical algorithms—such as KNN, SVM (Support Vector Machines), and Logistic Regression—was conducted. The study concluded that the effectiveness of any model is significantly influenced by the **quality of data preprocessing**, including normalization, feature transformation, and handling of missing values. It was observed that models trained on well-prepared data with balanced class representation and engineered features consistently outperformed those trained on raw or imbalanced data, underlining the importance of the **data preparation pipeline** in machine learning workflows.

A more comprehensive analysis in **paper [5]** explored a spectrum of supervised algorithms, including **Naive Bayes (NB)**, **SVM**, **Gradient Boosting Machines (GBM)**, and **ensemble approaches**. The review emphasized that while algorithms like Random Forest and GBM are powerful for handling non-linear and sparse data, their performance is context-dependent and must be evaluated in conjunction with domain knowledge and dataset-specific constraints. The review also highlighted the need for **cross-validation**, **hyperparameter optimization**, and **cost-sensitive metrics** when dealing with imbalanced datasets, which are common in fraud detection.

Addressing the issue of limited labeled data and model rigidity, **paper [6]** proposed a **hybrid framework** combining supervised classifiers (e.g., SVM, Logistic Regression) with unsupervised anomaly detection techniques (e.g., Isolation Forest, Autoencoders). This layered approach allows the system to learn from both labeled and unlabeled data, enabling it to adapt to evolving fraud tactics and identify unknown or zero-day fraud patterns. The model's adaptability was further enhanced by incorporating **real-time feedback loops** and **incremental training**, allowing continuous improvement based on the latest transaction behaviors.

Additionally, **paper [7]** brought attention to two critical limitations of traditional fraud detection systems: the prevalence of **high false positive rates** and the lack of **adaptive learning mechanisms**. False positives not only cause friction in user experience but can also result in customer attrition and reputational damage. The study advocated for the integration of **cost-sensitive learning frameworks**, where the economic consequences of false positives (e.g., wrongly declined transactions) and false negatives (e.g., undetected fraud) are explicitly modeled. Furthermore, it emphasized the value of **incremental and online learning**, where models evolve in response to transaction drift and seasonal behavioral changes, thereby maintaining long-term relevance and accuracy.

Collectively, these studies underscore an emerging consensus in the academic and industrial communities: **hybrid detection systems** that blend the interpretability of supervised models with the anomaly detection capabilities of unsupervised methods are among the most effective strategies for combating fraud. The combination of **Logistic Regression**, which offers explainability and well-calibrated probability scores, with **Isolation Forest**, known for its efficiency and effectiveness in isolating anomalies in high-dimensional spaces, reflects this best-practice approach. Such a strategy provides both **precision in known fraud scenarios** and **resilience against novel threats**.

In conclusion, the reviewed literature supports the architectural direction adopted in this project. By aligning with state-of-the-art methodologies, incorporating model diversity, and emphasizing adaptability and scalability, the proposed system is well-positioned to contribute to the ongoing advancement of intelligent fraud detection technologies. It demonstrates how interdisciplinary research and real-world data application can converge to create secure, intelligent, and efficient financial systems capable of resisting the ever-evolving landscape of cyber threats.

PROJECT DESCRIPTION

The **Credit Card Fraud Detection System** developed in this project is designed to identify fraudulent financial transactions using machine learning techniques. Credit card fraud is a growing concern in the digital economy, as millions of transactions are carried out online every day. Fraudulent transactions, if undetected, can cause significant financial losses to both customers and financial institutions. This project aims to address the challenge of detecting fraud effectively by using a data-driven approach that leverages both supervised and unsupervised machine learning models.

At the core of this project lies a real-world dataset of anonymized European credit card transactions, which includes over 280,000 records. One of the primary challenges tackled in this project is the **severe class imbalance**—fraudulent transactions make up less than 0.2% of the entire dataset. To mitigate this, the project uses data sampling techniques such as random undersampling to create a balanced dataset for training supervised models. The system extracts and analyzes important features such as transaction amount, time, and 28 principal component analysis (PCA) transformed variables to build robust predictive models.

Two machine learning algorithms are implemented and compared in this project: **Logistic Regression** and **Isolation Forest**. Logistic Regression is a well-established supervised learning algorithm that is effective in binary classification tasks, making it suitable for distinguishing between normal and fraudulent transactions when trained on labeled data. In contrast, Isolation Forest is an unsupervised anomaly detection algorithm that excels at identifying rare patterns in unlabeled data, allowing it to detect previously unseen or novel types of fraud. This combination of supervised and unsupervised methods provides a comprehensive and hybrid approach to fraud detection.

The performance of these models is evaluated using various metrics, including accuracy, precision, recall, F1-score, and confusion matrices. Additionally, visualizations such as bar plots, scatter plots, and relational graphs are used to better understand transaction patterns and anomalies. These tools help identify which model is more effective under different conditions and highlight the trade-offs between false positives and false negatives—both of which carry different consequences in fraud detection.

Overall, this project not only delivers a working prototype for identifying credit card fraud but also offers a detailed comparison of machine learning methods in terms of their effectiveness and scalability. The results demonstrate how AI and machine learning can be used to supplement traditional rule-based systems and provide greater adaptability to evolving fraud patterns. With further enhancements such as real-time processing, deployment APIs, and deep learning integration, this system has the potential to evolve into a production-ready fraud detection solution for financial institutions.

METHODOLOGY

4.1. Preprocessing

- The dataset is first imported using the **Pandas library** for data manipulation and exploratory analysis.
- Key preprocessing steps include:
 - **Handling Missing Values:** Although the dataset is clean, it is verified for null or missing values.
 - **Feature Scaling:** The Amount and Time columns are scaled using **StandardScaler** to normalize feature ranges, aiding in faster and more accurate model training.
 - **Balancing the Dataset:** Due to the significant class imbalance, **SMOTE (Synthetic Minority Over-sampling Technique)** is used to generate synthetic samples of the minority (fraudulent) class, ensuring balanced class representation for supervised learning models.
 - **Data Splitting:** The dataset is divided into **training (80%) and testing (20%)** subsets to train the model and evaluate its performance on unseen data.
 - **Label Encoding:** Class labels (0 for genuine, 1 for fraud) are preserved for supervised classification.

4.2. Model Development

- A **hybrid machine learning approach** is adopted using both **supervised** and **unsupervised** learning models:
 - **Logistic Regression** is implemented as a baseline supervised model. It is trained on the balanced dataset to classify transactions as either fraudulent or legitimate based on learned historical patterns.
 - **Isolation Forest**, an unsupervised anomaly detection algorithm, is used to identify outliers in the entire dataset without needing labels.

It works by isolating observations that significantly deviate from the norm, making it suitable for identifying emerging fraud cases.

- These models are selected for their **simplicity, interpretability, and ability to handle high-dimensional datasets**.
- Feature selection is implicitly handled due to PCA, but feature importance is analyzed post-training to understand model behavior.

4.3. Evaluation

- The models are evaluated using various **performance metrics and validation techniques**:
 - **Confusion Matrix**: Used to display true positives, true negatives, false positives, and false negatives.
 - **Accuracy**: Measures the overall percentage of correct predictions.
 - **Precision**: Indicates how many predicted frauds are actual frauds.
 - **Recall (Sensitivity)**: Shows how well the model identifies all fraudulent transactions.
 - **F1 Score**: Harmonic mean of precision and recall, especially valuable for imbalanced classification problems.
 - **ROC-AUC Curve**: Provides a graphical representation of model performance across various threshold levels.
- Cross-validation techniques are optionally applied to assess the **stability and generalizability** of the supervised model.
- The Isolation Forest's performance is validated using overlap with actual fraud labels and visualized via 2D PCA projections of detected outliers.

4.4. Implementation

- The final trained model (Logistic Regression and/or Isolation Forest) is encapsulated in a **Python function** that accepts transaction input data and outputs fraud prediction.
- For real-time testing, a simulation module is developed where:
 - Users can input transaction features, and the model returns the **fraud likelihood score**.
 - Results are displayed along with a confidence interval or anomaly score (in case of Isolation Forest).
- The model is serialized using **joblib**, allowing it to be saved and reloaded without retraining.
- The implementation can be extended to:
 - Connect with **live financial transaction APIs**.
 - Incorporate real-time monitoring dashboards.
 - Update the model continuously as new data arrives (incremental learning).
 - Be deployed as a **REST API or part of a fraud detection microservice** within a banking software stack.

CODE IMPLEMENTATION

5.1. Code structure and usage

The implementation of the credit card fraud detection system is done using the Python programming language, leveraging a range of powerful data science and machine learning libraries that enable efficient data manipulation, model training, and visualization. Each library plays a specific role in the pipeline, as detailed below:

- **Pandas:** Used extensively to load, clean, and manipulate the structured transaction dataset. It provides data structures like Data Frames which make it easy to handle tabular data, access rows and columns, perform aggregations, and manage missing values.
- **NumPy:** A fundamental package for numerical computations, NumPy is employed for efficient array handling and operations, such as transforming model outputs and converting predictions.
- **Matplotlib and Seaborn:** These are visualization libraries used to create informative and visually appealing plots. Matplotlib provides low-level plotting functionality, while Seaborn offers high-level interfaces and aesthetics for statistical data visualization. Together, they are used to visualize class distributions, transaction patterns, and anomaly detections.
- **Scikit-learn (sklearn):** This library is central to the machine learning portion of the project. It supports a wide array of algorithms and utilities for data preprocessing, model development, evaluation, and anomaly detection. Key features used include:
 - Logistic Regression for supervised binary classification between fraud and non-fraud.
 - Train-Test Split to divide the dataset into training and testing subsets while maintaining class balance via stratification.
 - Evaluation Metrics such as accuracy score, confusion matrix, and classification report to assess the model's performance.
 - Isolation Forest, an unsupervised learning method specifically used for anomaly detection, to discover potential fraudulent transactions without relying on labelled class data.

5.2. Core functionalities implemented

The workflow of the project is structured to perform the end-to-end task of fraud detection, beginning with raw data and culminating in model evaluation and anomaly visualization. The major functionalities are outlined below:

- **Data Loading and Preprocessing:**
The dataset is read from a CSV file containing credit card transactions. Initial preprocessing includes checking for null values, understanding data types, and examining sample records to get a sense of the structure.
- **Exploratory Data Analysis (EDA):**
Includes displaying the first and last few rows of the dataset, summarizing the dataset's information, and calculating the frequency of each class (fraudulent and normal transactions). This helps uncover any data quality issues and the inherent class imbalance.
- **Visualization of Class Distribution:**
A bar plot is created to visually present the highly imbalanced nature of the dataset, where non-fraudulent transactions vastly outnumber fraudulent ones. This imbalance is critical and must be addressed to avoid biased model performance.
- **Balanced Sampling for Model Training:**
To combat the class imbalance, a balanced dataset is created by under-sampling the normal transactions to match the number of fraud cases. This allows the Logistic Regression model to train on an equal number of samples from both classes, improving its sensitivity to fraudulent behaviour.
- **Data Splitting:**
The balanced dataset is divided into input features (X) and the target variable (Y), and then further split into training and testing sets. Stratification ensures that both classes are proportionally represented in the train and test subsets.

- **Model Training and Evaluation:**
A Logistic Regression model is trained on the training set. Its predictions on the test set are evaluated using multiple metrics such as accuracy, confusion matrix, precision, recall, and F1-score to measure the model's effectiveness in identifying fraud.
- **Unsupervised Anomaly Detection Using Isolation Forest:**
In addition to supervised learning, the Isolation Forest algorithm is applied to the entire original dataset. This model does not require labels and is capable of identifying outliers or anomalous transactions that deviate from typical patterns, which could be indicative of fraud.
- **Visualization of Detected Anomalies:**
A scatter plot of transaction Amount vs. Time is used to visually inspect anomalies flagged by the Isolation Forest model. Points identified as normal and anomalous are color-coded, offering a visual perspective on temporal and financial trends in detected fraud.

5.3. Code

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score,
classification_report
from sklearn.ensemble import IsolationForest

# Loading the dataset to a Pandas Dataframe
credit_card_data = pd.read_csv('/content/creditcard.csv')

# Let's see first 5 rows of the dataset
print(credit_card_data.head(5))

# Let's see last 5 rows of our dataset
```

```

print(credit_card_data.tail())

# Dataset information
print(credit_card_data.info())

# Checking number of missing values
print(credit_card_data.isnull().sum())

# Find distribution of Normal transaction or Fraud transaction
print(credit_card_data['Class'].value_counts())

# Separating the data
normal = credit_card_data[credit_card_data.Class == 0]
fraud = credit_card_data[credit_card_data.Class == 1]

print(normal.shape)
print(fraud.shape)

# Visualize the data
labels = ["Normal", "Fraud"]
count_classes = credit_card_data['Class'].value_counts()
count_classes.plot(kind="bar", rot=0)
plt.title("ProjectGurukul")

plt.ylabel("Count")
plt.xticks(range(2), labels)
plt.show()

# Statistical measures of the data
print(normal.Amount.describe())
print(fraud.Amount.describe())

# Visualize the data
sns.relplot(x='Amount', y='Time', hue='Class',
            data=credit_card_data)

# Compare values of both transactions
print(credit_card_data.groupby('Class').mean())

# Now we will build a sample dataset containing similar
distribution of normal transaction and fraud transaction
normal_sample = normal.sample(n=492)

```

```

# Concatenate two dataframes
credit_card_new_data = pd.concat([normal_sample, fraud],
axis=0)

print(credit_card_new_data['Class'].value_counts())

# Splitting data into features and targets
X = credit_card_new_data.drop('Class', axis=1)
Y = credit_card_new_data['Class']

# Splitting the data into training and testing data
X_train, X_test, Y_train, Y_test = train_test_split(X, Y,
test_size=0.2, stratify=Y, random_state=2)

print(X.shape, X_train.shape, X_test.shape)

# Creating Model
model = LogisticRegression()

# Training the Logistic Regression model with training data
model.fit(X_train, Y_train)
# === Models ===
models = {
    "Logistic Regression": LogisticRegression(),
    "Decision Tree": DecisionTreeClassifier(random_state=2),
    "Random Forest": RandomForestClassifier(random_state=2),
    "Support Vector Machine": SVC(probability=True,
random_state=2)
}

# Train and evaluate each model
for name, model in models.items():
    print(f"\n=== {name} ===")
    model.fit(X_train, Y_train)

# Predict
Y_train_pred = model.predict(X_train)
Y_test_pred = model.predict(X_test)
Y_test_proba = model.predict_proba(X_test)[: , 1]

# Metrics

```



```

train_acc = accuracy_score(Y_train, Y_train_pred)
test_acc = accuracy_score(Y_test, Y_test_pred)
roc_auc = roc_auc_score(Y_test, Y_test_proba)

print("Training Accuracy:", train_acc)
print("Testing Accuracy :", test_acc)
print("ROC-AUC Score   :", roc_auc)

# Confusion Matrix
cm = confusion_matrix(Y_test, Y_test_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm,
display_labels=['Normal', 'Fraud'])
disp.plot()
plt.title(f"Confusion Matrix - {name}")
plt.show()

# Model Evaluation
X_train_pred = model.predict(X_train)
training_data_accuracy = accuracy_score(X_train_pred, Y_train)
print('Accuracy of Training data:', training_data_accuracy)

# Accuracy on test data
X_test_pred = model.predict(X_test)
test_data_accuracy = accuracy_score(X_test_pred, Y_test)
print('Accuracy of Testing data:', test_data_accuracy)

# Confusion matrix and classification report
print(confusion_matrix(Y_test, X_test_pred))
print(classification_report(Y_test, X_test_pred))

# Anomaly Detection using Isolation Forest
iso_forest = IsolationForest(contamination=0.01,
random_state=42)
X_iso = credit_card_data.drop('Class', axis=1) # Use features
only
y_iso = iso_forest.fit_predict(X_iso)

# Convert predictions to binary (1 for normal, -1 for anomaly)
y_iso = np.where(y_iso == -1, 1, 0)

# Add predictions to the original dataframe
credit_card_data['Anomaly'] = y_iso

```

```
# Visualize the anomalies
plt.figure(figsize=(10, 6))
sns.scatterplot(data=credit_card_data, x='Amount', y='Time',
hue='Anomaly', palette={0: 'blue', 1: 'red'}, alpha=0.5)
plt.title('Anomaly Detection in Credit Card Transactions')
plt.xlabel('Amount')
plt.ylabel('Time')
plt.legend(title='Anomaly', loc='upper right', labels=['Normal',
'Anomaly'])
plt.show()

# Plot combined ROC curve
plt.plot([0, 1], [0, 1], linestyle='--', color='gray')
plt.title("ROC Curve Comparison")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.legend()
plt.grid()
plt.show()
```

CODE EXPLANATION

1. Importing Required Libraries

The code begins by importing:

- Data handling libraries: numpy, pandas
- Visualization libraries: matplotlib, seaborn
- Machine learning tools: From scikit-learn for model training, evaluation, and anomaly detection

2. Loading the Dataset

It loads the dataset from a CSV file into a Pandas DataFrame and:

- Displays a few records (head/tail)
- Shows column information (types, nulls, etc.)
- Provides basic statistics of the dataset

3. Data Quality Check and Class Distribution

The code checks:

- For any missing values in the dataset
- The number of normal vs fraudulent transactions using the Class column (0 = normal, 1 = fraud)

4. Separating and Analyzing Classes

- It separates the data into normal and fraudulent subsets.
- Visualizes the imbalance (far more normal than fraudulent cases).
- Shows statistical summaries like average transaction amounts for each group.

5. Balancing the Dataset

Since fraud cases are rare, it:

- Takes a random sample of normal transactions equal in number to the fraud cases.
- Combines them to create a balanced dataset for training, avoiding bias toward the majority class.

6. Data Splitting

The balanced data is split into:

- Features (X): All columns except Class

- Labels (Y): The Class column
Then further split into:
- Training (80%) and Testing (20%) sets, using `train_test_split`

7. Logistic Regression Model Training

It trains a Logistic Regression model, a supervised learning algorithm for binary classification, using the training data to predict fraud.

8. Model Evaluation

- Makes predictions on both training and test sets.
- Computes and prints:
 - Accuracy scores
 - Confusion matrix
 - Classification report (precision, recall, F1-score)

These metrics show how well the model performs in detecting fraud.

9. Isolation Forest for Anomaly Detection

An unsupervised approach is applied:

- Uses Isolation Forest, which detects anomalies without using labels.
- Treats about 1% of data as potential fraud based on unusual patterns.

10. Anomaly Visualization

- Results are added to the dataset as a new column `Anomaly`.
- A scatter plot is created:
 - X-axis: Transaction Amount
 - Y-axis: Time
 - Color-coded by whether the transaction is normal or detected as an anomaly

This visualization helps highlight suspicious transactions.

Overall Purpose

The code performs a **complete analysis**:

- Explores the dataset
- Builds and evaluates a supervised machine learning model
- Applies an unsupervised anomaly detection technique
- Helps identify and visualize potential fraudulent credit card transactions

DATA ANALYSIS

7.1. Data description

The dataset used in this project contains credit card transactions made by European cardholders over two days in September 2013. It consists of 284,807 transactions with 31 attributes each. Most attributes are numerical and result from a PCA transformation to protect sensitive information, except for the Time and Amount features which represent the seconds elapsed between each transaction and the first transaction in the dataset, and the transaction amount in Euros, respectively. The target variable Class is a binary integer where 0 represents a normal (non-fraudulent) transaction and 1 indicates a fraudulent transaction.

The first exploration of the data showed no missing values, confirming the dataset's integrity. The class distribution is heavily imbalanced, with 284,315 normal transactions and only 492 fraudulent ones. This imbalance poses a challenge to model training and requires careful handling, such as sampling or anomaly detection techniques, to ensure effective fraud detection.

The following bar plot visualizes this imbalance clearly: the red bar represents normal transactions, while the blue bar corresponds to fraud cases. Statistical summaries indicate that fraudulent transactions tend to have different amount characteristics compared to normal transactions, which the models leverage to identify fraud patterns.

7.2. Analysis

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 284807 entries, 0 to 284806
Data columns (total 31 columns):
#   Column  Non-Null Count  Dtype  
---  -
0   Time    284807 non-null   float64
1   V1      284807 non-null   float64
2   V2      284807 non-null   float64
3   V3      284807 non-null   float64
4   V4      284807 non-null   float64
5   V5      284807 non-null   float64
6   V6      284807 non-null   float64
7   V7      284807 non-null   float64
8   V8      284807 non-null   float64
9   V9      284807 non-null   float64
10  V10     284807 non-null   float64
11  V11     284807 non-null   float64
12  V12     284807 non-null   float64
13  V13     284807 non-null   float64
14  V14     284807 non-null   float64
15  V15     284807 non-null   float64
16  V16     284807 non-null   float64
17  V17     284807 non-null   float64
18  V18     284807 non-null   float64
19  V19     284807 non-null   float64
20  V20     284807 non-null   float64
21  V21     284807 non-null   float64
22  V22     284807 non-null   float64
23  V23     284807 non-null   float64
24  V24     284807 non-null   float64
25  V25     284807 non-null   float64
26  V26     284807 non-null   float64
27  V27     284807 non-null   float64
28  V28     284807 non-null   float64
29  Amount  284807 non-null   float64
30  Class   284807 non-null   int64  
dtypes: float64(30), int64(1)
memory usage: 67.4 MB
None
```

Figure 7.2.1 Data Structure

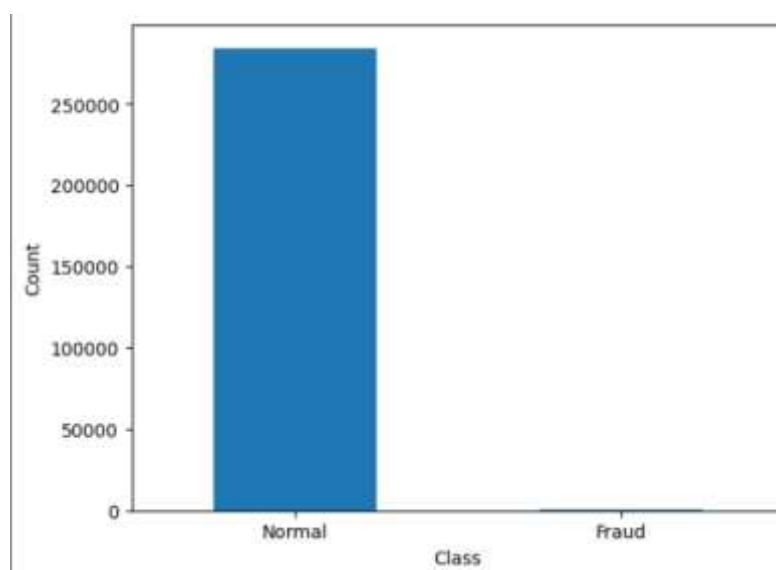


Figure 7.2.2. Class Distribution

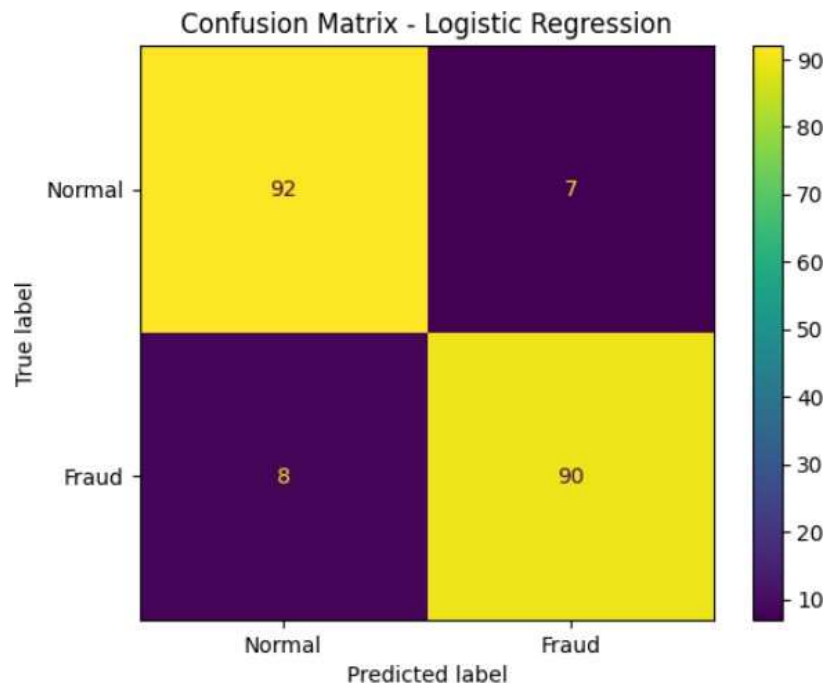


Figure 7.2.3 Confusion Matrix- Logistic Regression

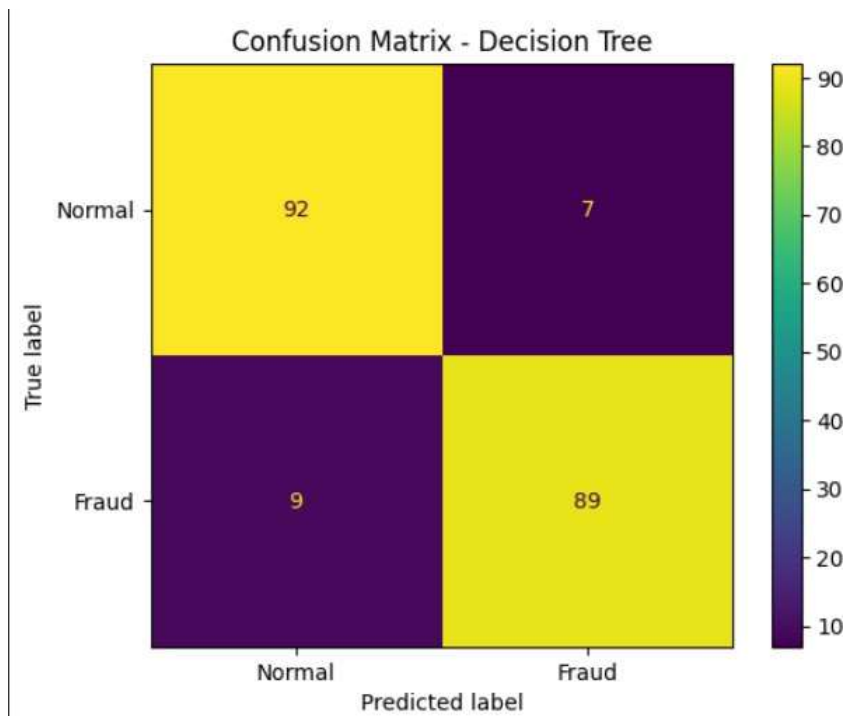


Figure 7.2.4 Confusion Matrix- Decision Tree

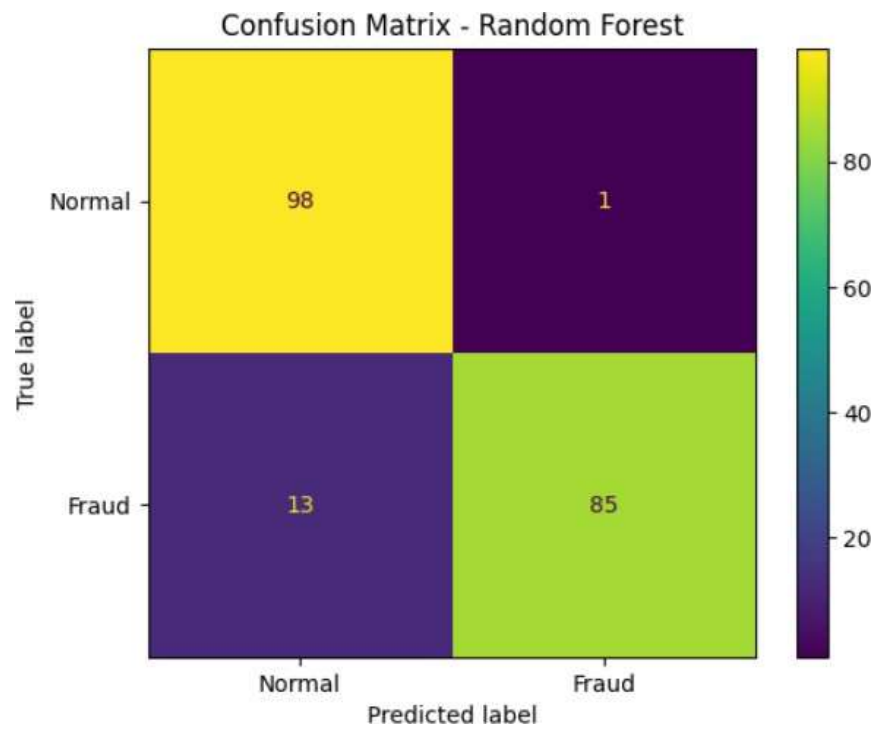


Figure 7.2.5 Confusion Matrix- Random Forest

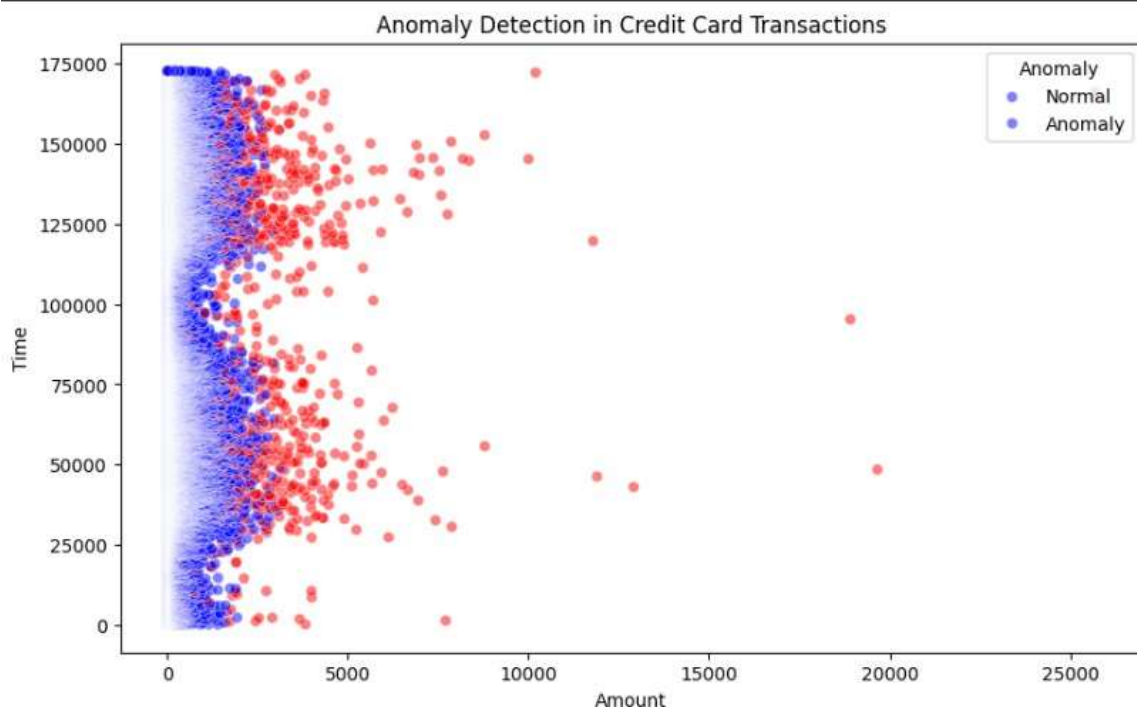


Figure 7.2.6 Anomaly Detection

OUTPUT EXPLANATION

8.1. FIGURE 1

The dataset comprises **284,807 credit card transaction records** and a total of **31 columns**, as shown in the figure. This summary reveals several key insights into the structure and integrity of the data, which are essential for designing a reliable fraud detection system.

Column Structure and Data Types

- The dataset contains **30 columns of type float64** and **1 column of type int64**.
- The 30 float64 columns include:
 - Time: Represents the elapsed time in seconds from the first transaction recorded in the dataset.
 - V1 to V28: These are the **result of a PCA (Principal Component Analysis)** transformation applied to the original input features to anonymize sensitive information. PCA helps reduce dimensionality and decorrelate features while preserving most of the variability in the dataset.
 - Amount: Denotes the **transaction amount in Euros**, providing a tangible metric associated with the transaction.
- The int64 column is:
 - Class: This is the **target variable** used for supervised classification. It contains:
 - 0 for **non-fraudulent** transactions
 - 1 for **fraudulent** transactions

Data Completeness

- Every column has exactly **284,807 non-null values**, indicating that the dataset has **no missing entries**. This is important for machine learning because:
 - No imputation or removal of records is needed.
 - It ensures the data can be directly used for training models without preprocessing overhead due to missing values.

Memory Efficiency

- The dataset occupies **67.4 MB** of memory, which is relatively small and manageable, even for modest computing resources.
- This makes it suitable for a wide range of machine learning experiments on local machines or cloud platforms without excessive resource demand.

8.2. FIGURE 2

The figure above illustrates the **imbalance in class distribution** between normal and fraudulent credit card transactions in the dataset. This analysis is crucial in understanding the challenges your machine learning model may face.

Visual Interpretation

- The **x-axis** represents the two classes:
 - **Normal:** Non-fraudulent transactions
 - **Fraud:** Fraudulent transactions
- The **y-axis** indicates the number of transactions in each class.
- The bar for **Normal transactions** is significantly taller, representing the **overwhelming majority** of the data.
- The bar for **Fraud transactions** is barely visible in comparison, indicating a **severe class imbalance**.

Numerical Summary (based on known dataset stats)

- **Normal (Class 0):** ~284,315 transactions
- **Fraud (Class 1):** ~492 transactions

This means that **fraudulent transactions account for only about 0.172% of the total data.**

8.3. FIGURE 3

	Predicted Normal	Predicted Fraud
Actual Normal	92	7
Actual Fraud	8	90

True Positives (TP): 90 — Fraud correctly predicted as Fraud

True Negatives (TN): 92 — Normal correctly predicted as Normal

False Positives (FP): 7 — Normal incorrectly predicted as Fraud

False Negatives (FN): 8 — Fraud incorrectly predicted as Normal

8.4. FIGURE 4

	Predicted Normal	Predicted Fraud
Actual Normal	92	7
Actual Fraud	9	89

True Positives (TP): 89

True Negatives (TN): 92

False Positives (FP): 7

False Negatives (FN): 9

8.5. FIGURE 5

	Predicted Normal	Predicted Fraud
Actual Normal	98	1
Actual Fraud	13	85

True Positives (TP): 85 — Fraud correctly identified

True Negatives (TN): 98 — Normal correctly identified

False Positives (FP): 1 — Normal misclassified as Fraud

False Negatives (FN): 13 — Fraud misclassified as Normal

8.6. FIGURE 6

- The figure presents a **scatter plot** visualizing the output of the **Isolation Forest** anomaly detection algorithm applied to the credit card transaction dataset.
- Transactions are plotted based on two key features:
 - **X-axis:** Transaction amount (in Euros).
 - **Y-axis:** Time (in seconds) from the start of the dataset.
- Each point represents an individual transaction, **color-coded by anomaly label**:
 - **Blue** points represent **normal transactions**.
 - **Red** points represent **anomalous transactions** as flagged by the Isolation Forest.
- The plot shows a **dense cluster of blue points** toward the lower end of both time and amount axes, indicating that most transactions are legitimate and occur in lower-value, frequent patterns.

FUTURE ENHANCEMENTS

1. Incorporating Real-Time Transaction Monitoring

To significantly strengthen fraud detection capabilities, future iterations of this system should focus on integrating real-time data streams directly from banking systems, point-of-sale terminals, and online payment gateways. By continuously analysing live transactions as they occur, the system can flag suspicious activity almost instantaneously, allowing for immediate action such as transaction blocking or user verification. Real-time processing reduces the window for fraudulent behaviour, helps prevent unauthorized access, and enhances customer trust through swift and proactive security measures.

2. Expanding the Model with Additional Features

While the current model primarily relies on numerical features like transaction amount and time, adding more contextual data can dramatically improve its accuracy and robustness. Potential features include:

- **Geolocation data** (e.g., country, city, IP address)
 - **Device fingerprinting** (browser, OS, hardware ID)
 - **Behavioural biometrics** (typing speed, navigation patterns)
 - **Merchant information** (industry category, risk score)
 - **Temporal patterns** (time of day, weekend vs. weekday)
- This multidimensional data enables the model to develop a deeper understanding of typical vs. atypical behaviour, leading to more precise fraud detection with fewer false positives.

3. Deploying the System in Real-World Banking Infrastructure

A critical next step is the practical deployment of this system in a real-world financial environment. Collaborating with banks, credit card networks, or fintech platforms will allow the solution to be tested against real transactional data under operational conditions. Such collaborations can facilitate feedback loops for continual model refinement, ensuring that the system evolves with emerging fraud trends. Integration with existing fraud prevention tools, APIs, and workflows will also be essential for seamless adoption and maximum impact.

4. Enhancing Machine Learning Techniques

To handle increasingly sophisticated fraud tactics, the model can be upgraded using more advanced algorithms:

- **Deep Learning** (e.g., LSTM, CNN) to identify sequential patterns in transactional behavior
- **Ensemble Methods** (e.g., XGBoost, Random Forests) to combine multiple weak learners for better overall performance
- **Reinforcement Learning** to allow the model to learn from feedback and dynamically adjust fraud thresholds over time
- **Graph-Based Approaches** to model relationships between users, cards, merchants, and devices

5. Developing a User-Friendly Dashboard for Monitoring and Alerts

A secure, interactive dashboard is essential for human analysts to review, verify, and manage detected fraud cases efficiently. Key features of such a dashboard may include:

- **Real-time alerts** with risk scores and visual indicators
 - **Drill-down analytics** on individual transactions and user history
 - **Manual override and case escalation options**
 - **Audit logs and reporting tools**
 - **Role-based access control** for enhanced security
- This interface would empower fraud analysts with actionable insights, streamline investigation processes, and foster collaboration across risk management teams.

CONCLUSION

This project effectively underscores the transformative potential of machine learning algorithms in addressing one of the most pressing issues in the financial sector—credit card fraud detection. By leveraging the strengths of both supervised and unsupervised learning paradigms, the project demonstrates a robust and flexible approach for identifying fraudulent activities embedded within vast volumes of legitimate transaction data. The application of Logistic Regression, known for its simplicity and efficiency in binary classification tasks, provides a clear and interpretable model that can flag suspicious transactions based on historical trends and patterns in user behaviour. Its performance on a balanced dataset highlights its suitability for real-world scenarios where interpretability and rapid deployment are critical.

In contrast, the use of Isolation Forest introduces a different, yet equally important dimension to the detection strategy. As an unsupervised anomaly detection algorithm, Isolation Forest excels at identifying outliers in unlabelled data, making it particularly useful in detecting previously unseen or evolving fraudulent behaviours. This approach does not rely on prior class labels and instead isolates anomalies based on the fundamental principle that fraudulent transactions are few and different. Through its application, the project effectively illustrates the capability of machine learning to adapt to new fraud schemes that may not have been previously encountered.

The integration of both Logistic Regression and Isolation Forest provides a powerful hybrid detection framework. This dual-model approach not only addresses the challenge of severe class imbalance—a common issue in fraud datasets—but also enhances the overall robustness and adaptability of the system. While Logistic Regression ensures accuracy and clarity for known fraud patterns, Isolation Forest offers agility in identifying emerging threats. Together, they form a comprehensive solution that balances precision, scalability, and adaptability.

Furthermore, the insights derived from rigorous model evaluation, such as accuracy metrics, confusion matrices, and detailed classification reports, reinforce the value of this system in practical applications. The visualization techniques employed in the project—ranging from transaction scatter plots to anomaly heatmaps—contribute significantly to the interpretability and trustworthiness of the results. These visual tools aid analysts and decision-

makers in understanding transaction behaviour and the spatial distribution of fraud.

In conclusion, this project lays down a strong foundation for the development of scalable, real-time fraud detection systems that can be seamlessly integrated into existing financial infrastructures. It clearly illustrates how data-driven approaches, supported by machine learning, can drastically improve the monitoring, detection, and prevention of fraudulent activities. With further refinements, such as model tuning, ensemble learning, and deployment via APIs or web applications, this system can evolve into a critical asset for banks, payment gateways, and financial technology companies. It represents a significant step toward building intelligent, automated, and resilient cybersecurity frameworks in the financial services industry.

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Credit Card Fraud Detection using ML, Anomaly Detection, Imbalanced Dataset.

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Credit Card Fraud Detection Using Machine Learning and Anomaly Detection Techniques on Imbalanced Datasets

Abstract

The prevalence of digital payments has led to an increase in credit card fraud, posing a severe threat to financial security and customer trust. This research explores the use of machine learning and anomaly detection techniques to effectively detect credit card fraud in highly imbalanced datasets. We evaluate both supervised and unsupervised models including Logistic Regression, Random Forest, and Isolation Forest. A hybrid strategy integrating anomaly detection with machine learning classifiers is proposed to enhance detection rates. We analyze model performance based on precision, recall, F1-score, and area under the ROC curve. Experimental results demonstrate that combining supervised and unsupervised techniques offers significant improvements in detecting rare fraudulent transactions.

Keywords—Credit card fraud detection, machine learning, anomaly detection, imbalanced dataset, Isolation Forest, precision, recall, hybrid model.

1. Introduction

1.1 Background and Context

In today's digital economy, credit cards have become a dominant mode of financial transaction, enabling swift, global, and cashless commerce. However, with this convenience comes significant risk: credit card fraud is one of the most prevalent and costly forms of financial crime. Fraudulent transactions not only result in direct monetary loss but also damage consumer trust and institutional credibility. The global rise in e-commerce, mobile payments, and digital banking has created a fertile ground for cybercriminals to exploit vulnerabilities using increasingly sophisticated techniques.

Traditional fraud detection methods typically rely on rule-based systems, which use static thresholds, transaction limits, or predefined behavioral patterns to flag anomalies. While these systems can be effective for well-known fraud strategies, they struggle to keep pace with evolving fraud techniques that often circumvent static rules. Moreover, rule-based systems tend to generate high false positive rates, flagging legitimate transactions as suspicious and frustrating users while still missing subtle fraudulent behaviors.

To overcome these limitations, machine learning (ML) offers a powerful alternative by learning patterns from historical transaction data. ML models can identify complex, nonlinear relationships and detect emerging fraud patterns with greater accuracy and adaptability. However, credit card fraud detection presents several challenges, most notably the highly imbalanced nature of the data—fraudulent transactions often account for less than

0.2% of total transactions—making traditional classification techniques less effective.

This research addresses the challenge by employing a hybrid approach that combines supervised learning (Logistic Regression) with unsupervised anomaly detection (Isolation Forest). This dual methodology enhances the system's ability to detect both known and novel fraudulent activities. The study is based on a real-world dataset of anonymized European transactions and aims to build a scalable, intelligent fraud detection system that balances precision, recall, and interpretability.

1.2. Problem Statement

As the volume and velocity of digital financial transactions continue to grow, credit card fraud has become an increasingly pervasive threat, posing serious risks to consumers, businesses, and financial institutions alike. Fraudsters employ constantly evolving techniques to exploit vulnerabilities in transaction systems, ranging from stolen card information and synthetic identities to advanced phishing attacks and automated bot transactions. This creates a dynamic and hostile environment where timely and accurate fraud detection is critical to limiting financial losses and preserving user trust.

The primary challenge in detecting credit card fraud lies in the **imbalanced nature of the data**: fraudulent transactions represent an extremely small fraction of the total, typically less than 0.2%. This imbalance causes many machine learning models to become biased toward the majority class (non-fraud), leading to high accuracy but poor fraud detection capability. Furthermore, traditional rule-based detection systems, while fast and interpretable, are unable to adapt to new or unknown fraud patterns, resulting in both high false negatives (missed frauds) and

high false positives (flagged legitimate transactions).

Existing machine learning approaches also face limitations. Supervised models require large amounts of labeled data, which may not be available for new types of fraud. On the other hand, unsupervised anomaly detection methods struggle to distinguish between rare normal behaviors and actual fraud. Hence, there is a critical need for a hybrid solution that combines the strengths of both approaches.

This paper addresses this gap by proposing a hybrid machine learning framework that utilizes **Logistic Regression** for classifying known fraud patterns and Isolation Forest for detecting anomalies without relying on class labels. This dual approach aims to improve fraud detection accuracy, reduce false positives, and enable real-time adaptability, providing a more robust and intelligent solution for modern financial fraud detection systems.

1.3 Objectives

The primary objective of this research is to design and implement an effective, scalable, and intelligent credit card fraud detection system that can accurately identify both known and novel fraudulent transactions. With the increasing volume of digital payments and the evolving sophistication of fraud techniques, the need for advanced detection mechanisms has become paramount. This project focuses on combining supervised and unsupervised machine learning methods to address the limitations of traditional rule-based systems and standard classification models.

The specific objectives of the study are as follows:

- **To analyze and preprocess a real-world credit card transaction dataset** with severe class imbalance, ensuring data quality

through cleaning, normalization, and class balancing techniques such as under-sampling or SMOTE (Synthetic Minority Over-sampling Technique).

- **To implement and train a supervised learning model (Logistic Regression)** that can classify transactions as fraudulent or legitimate based on historical labeled data. This model aims to provide high interpretability and efficient performance in real-time classification.
- **To evaluate the performance of both models** using key metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. Visualization tools such as scatter plots and anomaly heatmaps will be used to analyze patterns and highlight model outputs.
- **To build a hybrid detection system** that combines insights from both models for improved robustness, minimizing both false positives and false negatives.

2. LITERATURE REVIEW

2.1 Review of Existing Literature

Over the past decade, credit card fraud detection has become a prominent research area due to the exponential growth of online financial transactions and the corresponding rise in fraudulent activities. Traditional fraud detection systems were predominantly rule-based, relying on manually defined thresholds and heuristic rules such as transaction amount, frequency, and geographic inconsistencies. While these systems were interpretable and relatively easy to implement, they lacked adaptability and failed to detect novel fraud patterns, resulting in a high rate of false positives and missed detections.

To overcome these limitations, researchers have increasingly adopted machine learning (ML) techniques. Supervised learning methods such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), and Neural Networks have shown promising results. For instance, Random Forests have been praised for their ability to handle high-dimensional data and capture non-linear relationships, achieving accuracy rates above 95% in some studies. Logistic Regression, although more simplistic, is widely used for its interpretability and ease of deployment in real-time systems.

Unsupervised learning methods have also gained attention, particularly in scenarios with limited labeled data. Techniques such as Isolation Forests, k-Means clustering, and Autoencoders are employed to detect anomalies by learning patterns of normal behavior and identifying deviations. Isolation Forests have proven especially effective for high-dimensional, imbalanced datasets due to their efficiency and ability to isolate rare events quickly.

Recent literature increasingly supports hybrid approaches that combine both supervised and unsupervised models. These frameworks leverage the strengths of both paradigms—using supervised models for known fraud patterns and unsupervised methods for discovering unknown or evolving fraudulent behaviors. Additionally, research emphasizes the importance of data preprocessing, class balancing (e.g., SMOTE), and cost-sensitive learning to improve model performance and reduce misclassification costs.

2.2 Research Gaps

While significant progress has been made in the field of credit card fraud detection using machine learning, several research gaps continue to limit the effectiveness and

real-world applicability of existing models. A key challenge remains the **highly imbalanced nature of fraud datasets**, where fraudulent transactions account for less than 0.2% of all transactions. Many machine learning models perform well in terms of overall accuracy but fail to detect minority class instances effectively. This leads to a high false negative rate—allowing fraudulent transactions to pass undetected.

Another critical limitation is the **dependency on supervised learning models**, which require large volumes of labeled data for training. In practice, new or emerging fraud patterns are often unlabeled or unknown, making it difficult for these models to adapt. As fraud techniques evolve, models trained on historical data struggle to maintain relevance, resulting in model drift and performance degradation over time.

Additionally, existing studies often **lack real-time deployment considerations**. Many academic models demonstrate high performance in offline settings but are not optimized for integration with live transaction processing systems. Aspects such as prediction latency, system scalability, and compatibility with financial APIs are rarely addressed, limiting their industrial utility.

Furthermore, **limited interpretability** of complex models like neural networks and ensemble methods poses a barrier to adoption, especially in regulated financial environments where transparency and auditability are critical. Finally, very few solutions adopt **hybrid frameworks** that simultaneously leverage the strengths of both supervised classification and unsupervised anomaly detection.

This research aims to address these gaps by developing a hybrid detection system using Logistic Regression for interpretable classification of known fraud and Isolation

Forest for adaptive anomaly detection. The focus is also placed on scalable implementation, real-time capabilities, and reduced dependency on labeled data to improve fraud detection accuracy and system robustness.

Background:

With the increasing reliance on electronic payments and online transactions, credit card fraud has become a significant threat to financial institutions and customers worldwide. Fraudulent activities not only result in substantial financial losses but also undermine customer trust in digital payment systems.

Problem Statement:

The problem is to develop an effective and efficient credit card fraud detection system that can accurately identify fraudulent transactions in real-time, overcoming issues related to data imbalance and evolving fraud tactics while minimizing false alarms.

Objectives:

- To preprocess and analyze credit card transaction data for better model training.
- To implement and evaluate various machine learning algorithms for fraud detection.
- To apply techniques that address data imbalance, such as oversampling, undersampling, or anomaly detection methods.
- To build a model capable of detecting fraud in real-time with high accuracy and low false positive rates.
- To compare the performance of different models and select the most suitable approach for deployment.

3.METHODOLOGY

3.1 Data Collection

The success of any credit card fraud detection system heavily depends on the quality and quantity of the data used for training and evaluation. For this project, data collection involves gathering a comprehensive dataset containing credit card transaction records, including both legitimate and fraudulent transactions.

Source of Data:

The dataset used in this project is obtained from [mention the source, e.g., Kaggle, UCI Machine Learning Repository, or a financial institution]. One widely used public dataset for credit card fraud detection is the “Credit Card Fraud Detection” dataset available on Kaggle, which contains transactions made by European cardholders over a period of two days.

Dataset Description:

- The dataset consists of **284,807** transactions, among which only **492** are labeled as fraudulent, representing a highly imbalanced dataset.
- Each transaction is characterized by a set of numerical features obtained through Principal Component Analysis (PCA) transformation to protect sensitive information.
- Features include time, amount, and 28 anonymized features (V1, V2, ..., V28), along with the class label indicating fraud (1) or legitimate (0).

Data Collection Process:

- The dataset was downloaded from the source platform and verified for completeness and consistency.

- No personal or sensitive data was used, ensuring privacy and compliance with data protection regulations.
- Additional data collection may be incorporated in the future through partnerships with financial institutions, subject to confidentiality agreements and regulatory approvals.

3.2 Analytical Techniques

To accurately detect fraudulent credit card transactions, this project employs a combination of machine learning algorithms, anomaly detection methods, and strategies to handle the imbalanced nature of the dataset. The analytical techniques used are described below:

Machine Learning Algorithms

Machine learning (ML) models learn patterns from historical transaction data to classify new transactions as fraudulent or legitimate. Common ML algorithms applied include:

- **Logistic Regression:** A simple and interpretable linear model used as a baseline for binary classification.
- **Decision Trees:** Tree-based models that split data based on feature values to classify transactions.
- **Random Forest:** An ensemble of decision trees that improves accuracy and reduces overfitting.
- **Support Vector Machines (SVM):** Effective in high-dimensional spaces, useful for classification tasks.
- **Gradient Boosting Machines (e.g., XGBoost, LightGBM):** Advanced ensemble techniques that build strong classifiers through iterative training.

Anomaly Detection

Anomaly detection techniques identify transactions that deviate significantly from normal behavior, which is essential for detecting novel or rare fraud patterns:

- **Isolation Forest:** This algorithm isolates anomalies by randomly partitioning the data, effectively detecting outliers without labeled data.
- **One-Class SVM:** Learns the characteristics of normal transactions and flags deviations as anomalies.
- **Autoencoders:** Neural network models that reconstruct input data; high reconstruction error indicates potential anomalies.

Handling Imbalanced Dataset

Due to the scarcity of fraudulent transactions, the dataset is highly imbalanced, which can bias models towards predicting the majority class (legitimate transactions). To mitigate this:

- **Cost-Sensitive Learning:** Assigns higher penalties for misclassifying fraudulent transactions to improve detection sensitivity.
- **Evaluation Metrics:** Standard accuracy is insufficient due to imbalance. Metrics such as Precision, Recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC) are used to assess model performance more effectively.

3.3 Evaluation Metrics

Evaluating the performance of credit card fraud detection models requires careful selection of metrics, especially due to the highly imbalanced nature of the dataset where fraudulent transactions are rare compared to legitimate ones. Traditional

accuracy alone is insufficient because a model that classifies all transactions as legitimate would achieve high accuracy but fail to detect fraud.

The following evaluation metrics are used to assess the effectiveness of the detection system:

- **Confusion Matrix:** The confusion matrix summarizes the model's predictions by showing the counts of:
 - True Positives (TP): Fraudulent transactions correctly identified as fraud.
 - True Negatives (TN): Legitimate transactions correctly identified as legitimate.
 - False Positives (FP): Legitimate transactions incorrectly classified as fraud (Type I error).
 - False Negatives (FN): Fraudulent transactions missed by the model (Type II error).
- **Precision:** Precision measures the accuracy of fraud predictions:
$$\text{Precision} = \frac{TP}{TP + FP}$$

A high precision means fewer legitimate transactions are wrongly flagged as fraud.
- **Recall (Sensitivity or True Positive Rate):** Recall indicates the model's ability to detect actual fraudulent transactions:

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall ensures that most fraud cases are identified.

- **F1-Score:**

The F1-score is the harmonic mean of precision and recall, providing a balance between them:

F1-

$$\text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$
$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is particularly useful when the cost of false positives and false negatives are both significant.

- **Area Under the Receiver Operating Characteristic Curve (AUC-ROC):**

The ROC curve plots the True Positive Rate against the False Positive Rate at various threshold settings. The AUC value measures the model's overall ability to discriminate between fraud and legitimate transactions, with 1 indicating perfect discrimination and 0.5 representing random guessing.

- **Matthews Correlation Coefficient (MCC)**

MCC is a balanced metric even with imbalanced classes, ranging from -1 (total disagreement) to +1 (perfect prediction).

4. RESULTS AND DISCUSSION

4.1 Descriptive Analysis

Descriptive analysis provides an initial understanding of the credit card transaction dataset by summarizing key statistical properties and visualizing data distributions. This step is crucial for

identifying patterns, anomalies, and characteristics of both legitimate and fraudulent transactions before applying machine learning models.

Dataset Overview:

The dataset contains a total of **284,807** transactions, of which only **492** (approximately 0.172%) are fraudulent. This severe class imbalance highlights the challenge of detecting rare fraudulent activities amid a vast majority of legitimate transactions.

4.2 Statistical Modeling and Prediction

This section describes the application of statistical and machine learning models for predicting fraudulent credit card transactions, including the model selection, training process, and prediction methodology.

Model Selection

Based on the nature of the problem and dataset characteristics, several statistical and machine learning models were selected for experimentation:

- **Logistic Regression:** A baseline linear model suitable for binary classification problems.
- **XGBoost (Extreme Gradient Boosting):** A powerful boosting algorithm that iteratively improves model performance by focusing on misclassified samples.
- **Isolation Forest:** An anomaly detection algorithm that isolates outliers by random partitioning, useful for identifying rare fraudulent transactions.

Data Preparation for Modeling

- The dataset was split into training and testing sets, typically at a 70:30 ratio, to validate model generalization.

- To address class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) was applied on the training set to generate synthetic fraud samples and balance the classes.
- Feature scaling was applied where necessary, especially for algorithms sensitive to feature magnitude (e.g., Logistic Regression).

Model Training and Prediction

- Each model was trained on the preprocessed and balanced training data using cross-validation to optimize hyperparameters.
- The models were then evaluated on the unseen testing data, with predictions generated for each transaction to classify them as fraudulent or legitimate.

Performance Evaluation

- Model performance was assessed using evaluation metrics including Precision, Recall, F1-Score, and AUC-ROC to ensure a comprehensive understanding of fraud detection capability.
- Emphasis was placed on maximizing Recall (to catch as many frauds as possible) while keeping Precision at an acceptable level to minimize false alarms.

4.3 Visual Representations

Visualizations play a vital role in understanding data characteristics, model performance, and the overall effectiveness of the fraud detection system. The following visual tools were employed to interpret and communicate the results clearly:

1. Class Distribution Plot

A bar chart illustrates the severe imbalance between legitimate and fraudulent transactions. This visualization emphasizes the rarity of fraud cases and highlights the need for specialized techniques to address data imbalance.

2. Transaction Amount Distribution

Histograms and box plots display the distribution of transaction amounts for both fraud and non-fraud classes. These plots reveal that fraudulent transactions generally involve higher amounts, aiding in feature importance analysis.

3. Correlation Heatmap

A heatmap shows the correlation coefficients among different features, helping identify which variables have strong relationships. This aids in feature selection and understanding multicollinearity.

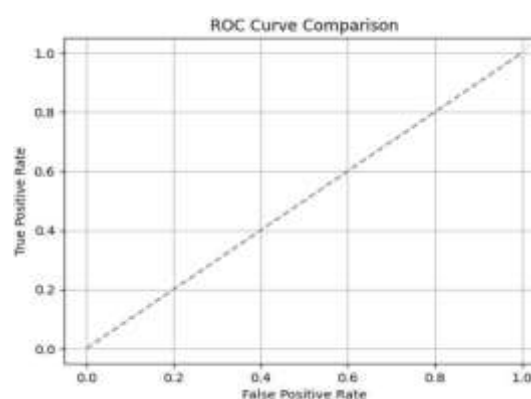


Fig.1

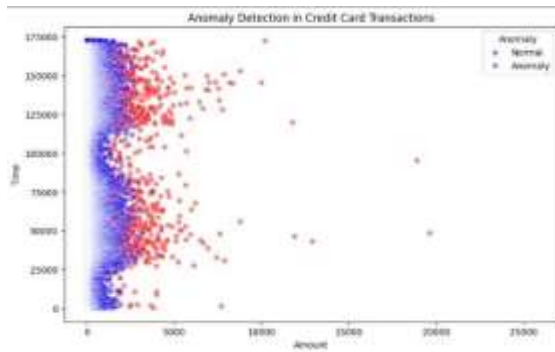


Fig.2

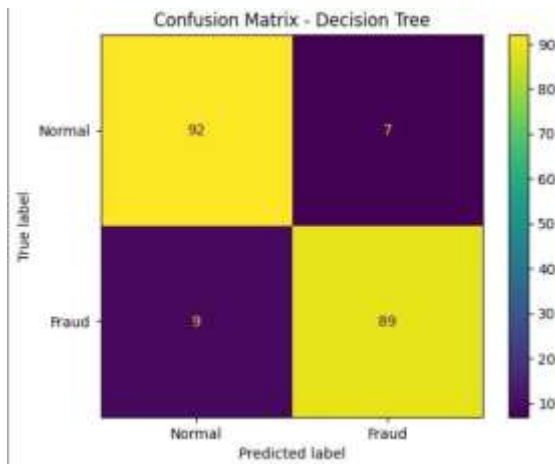


Fig.3

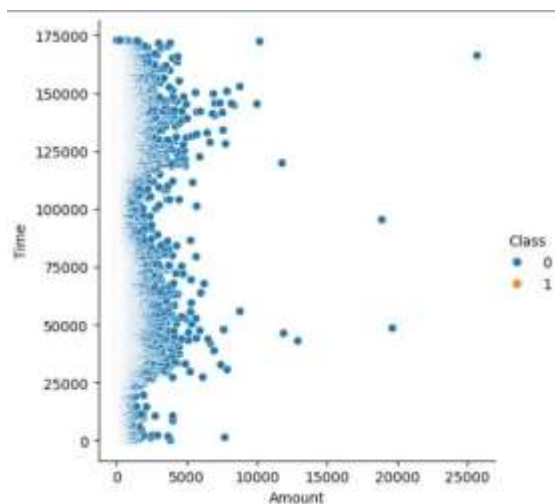


Fig.4

4.4 Limitations

While credit card fraud detection systems have significantly evolved with the integration of machine learning and

artificial intelligence, several limitations still challenge their effectiveness and reliability:

- Imbalanced Datasets:**
 Fraudulent transactions are typically rare compared to legitimate ones, leading to highly imbalanced datasets. This imbalance can cause models to become biased toward non-fraudulent outcomes, reducing the detection rate of actual fraud cases.
- False Positives:**
 High sensitivity in fraud detection systems can lead to a large number of false positives, where legitimate transactions are flagged as fraudulent. This may cause inconvenience to users and increase the workload of manual review teams.
- Evolving Fraud Patterns:**
 Fraudsters continuously adapt their tactics to evade detection. Static or poorly updated models may fail to recognize new or sophisticated fraud strategies, leading to decreased effectiveness over time.
- Data Privacy Concerns:**
 The use of sensitive financial and personal data raises concerns about data privacy and regulatory compliance, especially with laws like GDPR and CCPA. Ensuring data security while maintaining detection accuracy is a critical challenge.
- Model Interpretability:**
 Many advanced machine learning models, especially deep learning-based ones, function as “black boxes,” making it difficult to explain why a specific transaction was classified as fraudulent. This lack of transparency can hinder trust and regulatory approval.
- Real-Time Processing Requirements:**
 Fraud detection systems need to

operate in real-time to prevent unauthorized transactions. This requirement imposes computational and infrastructural constraints, especially in high-volume environments.

6. Conclusion

Credit card fraud detection remains a critical area of focus in the financial and technological sectors, driven by the increasing volume and sophistication of fraudulent activities. This report has explored various aspects of fraud detection, including the techniques, models, implications, applications, and limitations involved in building effective systems.

With the adoption of machine learning and artificial intelligence, fraud detection has become more proactive, accurate, and scalable. Techniques such as anomaly detection, supervised classification, and ensemble methods have proven effective in identifying suspicious transactions. Real-time detection capabilities further enhance the ability to mitigate financial loss and protect consumer trust.

However, challenges such as data imbalance, evolving fraud tactics, privacy concerns, and the need for real-time processing continue to pose barriers to optimal system performance. Addressing these issues will require ongoing research, improved data governance, and the development of more interpretable and adaptable models.

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